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Int. MSc. Examination – June 2025

Second semster

Int. MSc. Mathematics with minor in Data Science &

Int. MSc. Physics with minor in Scientific Computing

24MAT120 Multivariable Calculus

Maximum marks : 100 Time : 3 Hours

Scientific calculators are allowed.

Qn. No.	Questions	Marks	CO	Blooms taxonomy level
1	Find the equation of the tangent plane and normal line at the point $(1,-1,-2)$ on the surface $2xz^2 - 3xy - 4x = 7$.	10	CO2	L3
2	Integrate the function $f(x,y,z) = (1+2xz)y$ through the parabola $z = y^2$ on the plane $x = 2$ from $(2,1,1)$ to $(2,2,4)$.	10	CO2	L3
3	Let $f(x,y) = \frac{(x-y)}{(x+y)}$. Find the directions u for which 1. $D_u f\left(-\frac{1}{2},\frac{3}{2}\right)$ is largest 2. $D_u f\left(-\frac{1}{2},\frac{3}{2}\right)$ is smallest	10	CO1	L3
4	Find the area of the region cut from the plane $x+2y+2z=5$ by the cylinder whose walls are $x = y^2$ and $x = 2-y^2$.	10	CO2	L3
5	Find $\frac{\partial h}{\partial x}$ and $\frac{\partial h}{\partial y}$ where $h(x,y) = f(u(x,y),v(x,y))$ and $f(u,v) = \frac{u^2+v^2}{u^2-v^2}$, $u(x,y) = e^{-x-y}$ $v(x,y) = e^{xy}$.	10	CO2	L3
6	A fluid's velocity field is given by $F = (e^x \cos y + yz)i + (xz - e^x \sin y)j + (xy + z)k$. Show that the flow from $(0,1,0)$ to $(1,\frac{1}{\sqrt{2}},1)$ is same along the curves $C_1 : t^2 i + \frac{1}{\sqrt{1+t^2}} j + \sqrt{t}k$ and $C_2 : t\sqrt{2-t^2}i + \frac{1}{\sqrt{1+t^2}} j + t\cos(1-t)k$.	10	CO3	L3

7	Evaluate the following integrals: $a)\int_{-a}^{a}\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dydx \qquad b)\int_{0}^{a}\int_{y}^{a}\frac{x^2}{\sqrt{x^2+y^2}}\,dxdy$	10	CO3	L3
8	Find the volume of the tetrahedron whose vertices are at $(0,0,0),(0,0,3),(3,0,3)$ and $(0,3,3)$.	10	CO3	L3
9	Verify Green's theorem for the vector field $F=(xy^2+x^2)i+(4x-1)j$ over the triangular region with vertices $(0,0),(0,3),(-3,0)$ in the counter clock wise direction.	10	CO4	L3
10	A moving body started from the point $(0,1,3)$ with initial velocity $v(0)=j$, has the acceleration vector $a(t)=2i+e^t j$ at time t. Find the position and speed of the particle at time $t=10$.	10	CO3	L3

Bloom's taxonomy levels:

L1 - Remember, L2 - Understand, L3 - Apply, L4 - Analyze, L5 - Evaluate, L6 - Create